Who Leaves Teaching, and Why?

The Profile of Teacher Attrition in the United States, 2005–2025

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Abstract

Who leaves teaching, and why? The national answers disagree by a factor of two. I build the first national dataset that can arbitrate, twenty years of teacher panels linked from Current Population Survey interviews that follow 54,944 teachers into whatever they do next, and I make three contributions. First, I reconcile the literature with itself, reproducing the official follow-up survey and the retrospective estimates from this single dataset once each source's instrument is imposed, so the disagreement is measurement rather than substance. Second, I provide the first national estimates of where leavers go and of what predicts each exit route, and they show that the twenty-year rise in attrition is a reallocation toward other jobs, a third of which remain in education, rather than an exodus from work, with opportunity driving the job-change routes and family driving the labor-force ones. Third, running the identical design across professions, I find that a new baby raises a teacher's probability of leaving her profession by six percentage points, the largest effect among four comparable professions and one that nursing does not show at all. The profession with the family-friendly reputation is the one that loses its new mothers.

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1. Introduction

Who leaves teaching, and why? Few questions in the economics of education carry more policy weight. Teachers are the main school input in the production of achievement (Rivkin, Hanushek and Kain, 2005; Chetty, Friedman and Rockoff, 2014), replacing a leaver is expensive and disruptive for the students left behind (Ingersoll, 2001; Ronfeldt, Loeb and Wyckoff, 2013), and the fear of a post-pandemic exodus has moved retention to the center of the education policy debate. Yet the national evidence disagrees with itself, since published annual attrition rates range from 6 to 18 percent depending on the source. In this paper I answer the question for the United States with twenty years of nationally representative panel data, following every teacher the Current Population Survey (CPS) observes between 2005 and 2025 for twelve months, and I show that the apparently contradictory numbers of this literature describe one phenomenon measured with different instruments.

What is known so far comes from three kinds of sources, each with a blind spot. The Teacher Follow-up Survey provides the official national attrition rate, 8 percent of public school teachers in 2021–22, but it runs only every few years, relies on self-reports, and observes leavers in a single retrospective questionnaire (NCES, 2024). State administrative records support the sharpest causal work, from the Texas panels of Hanushek, Kain and Rivkin (2004) to the Wisconsin pay reform of Biasi (2021) and the post-pandemic studies of Massachusetts, North Carolina and Arkansas (Bacher-Hicks, Chi and Orellana, 2023; Bastian and Fuller, 2023; Goldhaber and Theobald, 2023; Camp, Zamarro and McGee, 2025), but each covers one state and loses sight of a teacher the moment she leaves the state payroll, so it cannot say where leavers go. The existing national estimates from the CPS, finally, rely on the March supplement question about the job held one year earlier (Harris and Adams, 2007; Aldeman and Yi, 2025), a retrospective self-report collected in a single interview that yields annual leaving rates of 6 to 8 percent but is known to understate occupational change and cannot observe returns, a gap I replicate and dissect in Appendix A.

I take a different route through the same survey. The CPS interviews each household in two blocks of four months separated by eight, so an adult observed in year t can be re-interviewed in the same calendar month of $t+1$. I assemble all 251 monthly files between January 2005 and December 2025, link the interviews twelve months apart through household and person identifiers with demographic validation in the spirit of Madrian and Lefgren (2000), and obtain twenty annual panels with 174,872 linked observations of 54,944 distinct teachers. Because one in six apparent leavers is teaching again within three months, I measure attrition with a strict definition that counts as a leaver only a teacher who is out of teaching at the twelve-month link and in every re-interview after it. The design observes the destination occupation and labor force status of every leaver, carries the full set of covariates of the survey, from fertility events to union status and pay, and updates every month at essentially no cost.

Five findings summarize the paper. First, the disagreement in published attrition rates is measurement, not substance. Within the same survey I reproduce the retrospective 6 to 8 percent family exactly, and a ladder of documented design differences, universe, substitutes, temporary returns and destination coding, connects my linked-design rate of 17.6 percent to a bracket of 8.0 to 11.5 that contains the official benchmark, so every published number fits inside one framework. Second, attrition is high and rising under the linked design, although not exceptional, since registered nurses leave their field faster in every year of the sample and the retrospective instrument, which I replicate profession by profession, places teachers below social workers and level with nurses and accountants. Every year 15.1 percent of American teachers leave the profession without an observed return, up from about 13 percent in the mid 2000s to 17 percent in recent years, with a new high of 18.0 percent in 2024. Third, the rise is a reallocation rather than a withdrawal. Moves to other occupations account for almost all of it, exits into non-participation are flat, and 30 percent of leavers remain inside education in jobs other than classroom teaching. Fourth, attachment to the job, not demographics, predicts who leaves. Part-time teachers are 11.3 percentage points more likely to leave than full-time colleagues and private-school teachers 7.5 points more likely than public-school ones, against a baseline of 15 percent, while higher pay retains. Fertility operates only through women, since a new baby raises mothers' exit probability by 5.1 points and leaves fathers' unchanged, the national counterpart, half a century later, of the classic cohort evidence in Stinebrickner (2002), and the same event estimated in comparison professions shows that this motherhood exit is a teaching phenomenon, absent among registered nurses. Fifth, geography lines up with what schools produce, since states with higher NAEP scores retain their teachers better.

The paper makes three contributions to the United States evidence. The first is to reconcile the literature with itself. To my knowledge this is the first study to nest the official follow-up survey, the retrospective estimates and a linked panel within a single dataset and to explain their disagreement, and in doing so it delivers the longest, most frequent and most current nationally representative attrition series, the first to purge temporary exits with a re-interview window. The second is destinations and routes. Administrative studies cannot observe a teacher once she leaves the public payroll and the follow-up surveys observe leavers only in survey years, whereas here every leaver carries an occupation code and a labor force status twelve months after baseline, so I can estimate not only who leaves but through which door. The third, and the sharpest new fact of the paper, is the first profession-by-profession estimate of the motherhood exit. Running the identical linked design on registered nurses, social workers and accountants shows that a new baby pushes teachers out of their profession like no comparable profession, six percentage points per birth, while nursing loses none of its new mothers, evidence that the family friendliness of teaching is reputation rather than fact. Around these contributions the paper estimates the broadest set of correlates yet fitted jointly on national data, including the event of a birth as distinct from the stock of children, union membership, sector, hours and pay, in one model with year fixed effects. None of the estimates is causal, and I offer them as the descriptive infrastructure that the causal literature, which is overwhelmingly single-state, currently lacks at the national level (Nguyen, Pham, Crouch and Springer, 2020).

The rest of the paper proceeds as follows. Section 2 describes the data and the leaver definition. Section 3 draws the portrait of the teaching workforce and its evolution. Section 4 documents attrition over time and Section 5 where leavers go. Section 6 asks who leaves and Section 7 why. Section 8 profiles the typical leaver and Section 9 concludes with the policy implications.

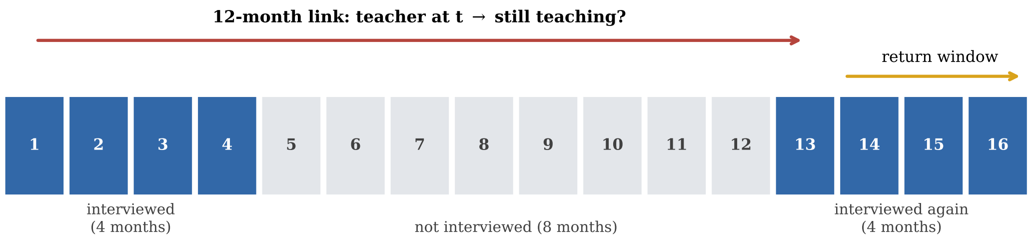
2. Data

I use the basic monthly microdata of the Current Population Survey (CPS), the household survey behind the official employment statistics of the United States, which I assemble for every month between January 2005 and December 2025.[2] What makes the CPS useful for studying attrition is its rotation design, illustrated in Figure 1. Each household is interviewed during four consecutive months, rests for eight, and is interviewed again during four more, so that any person can be observed for at most sixteen months and only once in a lifetime. Within that window, an adult interviewed in year t is re-interviewed in the same calendar month of $t+1$, and I link the two interviews through household and person identifiers, validating

[2] The files come from the U.S. Census Bureau for 2024–2025 and from the NBER mirror for 2005–2023. October 2025 was never released because the federal government shutdown suspended that month's data collection.

every match on sex, race and a plausible age progression (Madrian and Lefgren, 2000). Linking the public CPS files across months is the standard way to use the survey's panel dimension, since the Census Bureau does not release a longitudinal person identifier; the official labor force flows of the Bureau of Labor Statistics and the longitudinal keys distributed by IPUMS are constructed the same way (Rivera Drew, Flood and Warren, 2014). Because nobody is followed across several years, each base year forms a fresh cohort, and I compute every rate in this paper wave by wave over twenty waves.

Figure 1: The CPS follows each person for at most 16 months.



Notes: Interview calendar of one rotation group; numbers are months since the first interview. The 12-month link compares an interview from the first block with the interview held exactly one year later, and the return window uses the remaining interviews of the second block.

I identify teachers by the occupation of their main job, which covers classroom teaching from preschool and kindergarten through elementary, middle and secondary school, together with special education,[3] and I require at least a bachelor's degree, the standard restriction in the attrition literature. Pooling the twenty waves delivers 174,872 linked observations of 54,944 distinct teachers, representing the 4.6 million teachers of the country; a person interviewed in the four months of her first block contributes up to four linked baselines, and standard errors are clustered by household throughout, which nests the person. A leaver could simply be anyone not employed as a school teacher twelve months after baseline, which is the convention of the NCES Teacher Follow-up Survey (NCES, 2024), but temporary exits such as parental leave or a short unemployment spell inflate that measure, and indeed 15.7% of such leavers are teaching again within three months. My main outcome is therefore the non-returning leaver, a teacher who is out of teaching at $t+12$ and in every later re-interview, computed on the 129,897 baseline observations, from 52,238 distinct teachers, whose rotation allows at least one re-interview. Appendix A traces the full population funnel, shows that linkage failure does not select against teachers, reconciles my rates with the published benchmarks, and discusses the measurement caveats that make them upper bounds on permanent exit.

3. The typical teacher

Before asking who leaves, it helps to know who teaches. Table 1 draws that portrait on every person I observe in a school-teaching occupation over the two decades, 116,649 individuals who represent about 5.2 million teachers in an average month, split by sex and set against the rest of the college-educated workforce. The typical teacher is a married woman in her early forties with children at home, and she

[3]These are Census occupation codes 2300 to 2330, where 2300 designates preschool and kindergarten teachers, 2310 elementary and middle school teachers, 2320 secondary school teachers and 2330 special education teachers. The codes are identical in the 2002, 2010 and 2020 occupational classifications used over the period, which keeps the series comparable for twenty-one years.

differs from other graduates on almost every margin the survey measures. Teaching is overwhelmingly female, at 78 against 47 percent, disproportionately white, and markedly less open to Asian and foreign-born graduates. Credentials cluster in the middle, since 88 percent of teachers hold at least a bachelor's degree and 45 percent a master's, ten points above other graduates, while doctorates remain rare. Almost three quarters work for the public sector against 15 percent elsewhere, and 45 percent are union members against 9 percent. Hours are standard full-time, and the price of this stability is pay, since the median teacher earns $1,154 a week, roughly a quarter less than comparable college-educated workers.

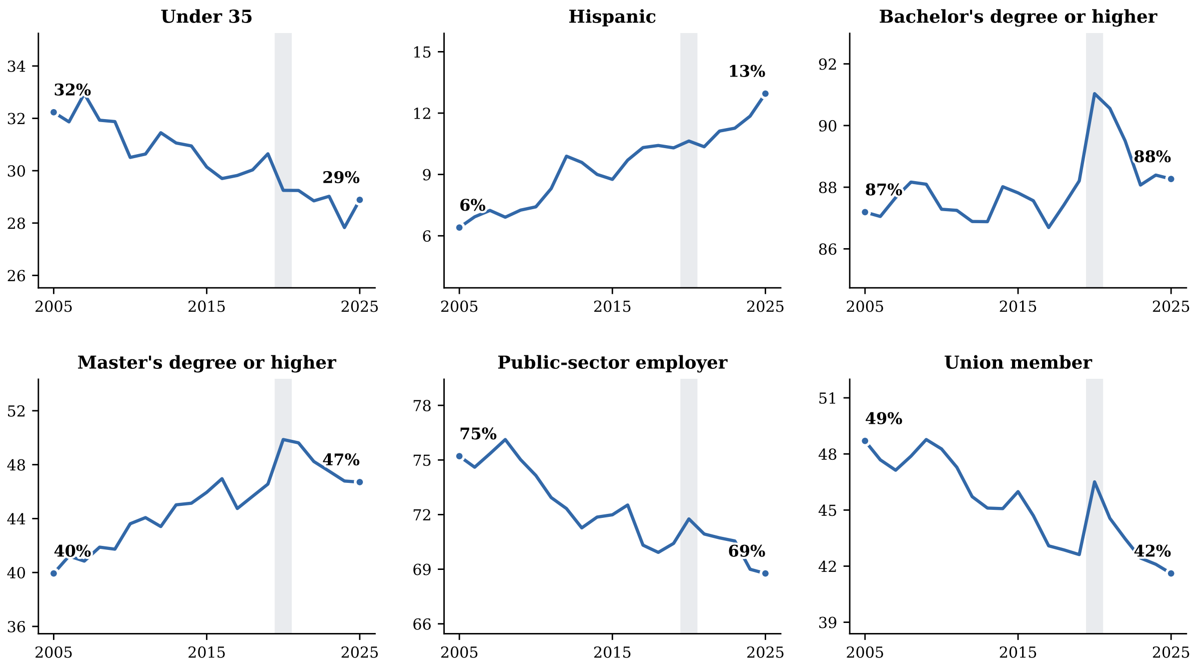
Table 1: Portrait of the American teaching workforce, 2005–2025.

	School teachers			Other college-educated workers	Difference
	Women	Men	All		
Panel A. Demographics					
Age, years	42.6	43.1	42.7	43.8	-1.11***
Female			78.2%	47.3%	30.9***
White	84.8%	85.3%	84.9%	78.5%	6.4***
Black	10.3%	9.8%	10.2%	9.0%	1.2***
Asian	2.8%	2.6%	2.8%	10.3%	-7.5***
Hispanic	9.4%	9.3%	9.4%	8.3%	1.0***
Non-citizen	2.2%	2.3%	2.3%	7.2%	-4.9***
Panel B. Family					
Married	65.8%	69.1%	66.6%	63.8%	2.7***
Number of own children (<18) at home	0.8	0.8	0.8	0.6	0.15***
Child under 6 at home	16.7%	19.9%	17.4%	15.6%	1.8***
Panel C. Education					
Bachelor's degree or higher	86.2%	94.4%	88.0%	100.0%	-12.0***
Master's degree or higher	43.5%	50.8%	45.1%	34.5%	10.6***
Professional degree or doctorate	2.1%	3.9%	2.5%	10.5%	-8.0***
Panel D. The job					
Public-sector employer	70.5%	78.0%	72.1%	15.4%	56.8***
Union member[a]	43.9%	50.2%	45.3%	8.7%	36.6***
Usual weekly hours	39.5	41.4	39.9	40.4	-0.49***
Part-time (<35 h/week)	11.7%	6.8%	10.6%	12.3%	-1.7***
Holds more than one job	6.4%	11.3%	7.4%	6.1%	1.3***
Panel E. Pay and income					
Weekly earnings, median[b]	$1,115	$1,308	$1,154	$1,500	−$346***
Family income $75k+[b]	81.1%	84.7%	81.9%	85.5%	-3.6***
Persons	92,044	24,812	116,649	963,782	
Population represented, millions[c]	4.1	1.1	5.2	49.6	

Notes: Weighted means over every employed person observed in a school-teaching occupation, with no degree restriction, pooling all 251 survey months; the comparison group is employed persons outside teaching holding at least a bachelor's degree. Sample sizes are unique persons, and a small number of persons with inconsistent sex codes across interviews appear in both sex columns. The last column reports the difference between all teachers and the comparison group, in percentage points for shares, and the significance of a two-sample test with standard errors clustered by household ($^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$). [a] Union membership and weekly earnings are only asked in the outgoing rotations, months in sample 4 and 8. [b] Measured over 2021–2025 only, to avoid mixing twenty years of price levels. [c] Population represented by the group in an average month. Source: author's calculations from CPS basic monthly microdata, 2005–2025.

The sex split carries its own message, because the minority of men who teach look systematically more attached to the core of the job, more often public-sector, unionized, full-time and holding a master's degree, and 11 percent of them hold a second job against 6 percent of women. The full age distributions

Figure 2: The changing profile of the teaching workforce.



Notes: Weighted annual means over all employed persons observed in a school-teaching occupation, with no degree restriction. Union membership is only asked in the outgoing rotations. The shaded band marks 2020. Appendix Table A5 reports the underlying numbers by sub-period with tests of the change. Source: author's calculations from CPS basic monthly microdata, 2005–2025.

and a visual summary of these contrasts are in Appendix Figures A1 and A2.

The profile has also been moving, and Figure 2 traces that movement year by year. The workforce is aging, with the share under 35 down from 32 to 29 percent, becoming markedly more Hispanic, from 6 to 13 percent, and steadily more credentialed, with bachelor's degrees edging up and master's degrees rising from 40 to 47 percent, while two anchors of the traditional teaching career erode at the same time, since public-sector employment falls from 75 to 69 percent of teachers and union membership from 49 to 42 percent. The gender mix, in contrast, has not moved at all. Teaching is drifting toward the private, non-union segment where, as the results below show, careers break most often. Appendix Table A5 reports the same evolution by sub-period together with formal tests, and every one of these changes is significant at the one percent level.

Weighted to population, the stock of degree-holding teachers, the group my attrition analysis follows, is remarkably stable at about 4.7 million over the whole period (Appendix Figure A3), with the composition by teaching level equally steady and dominated by elementary and middle school (shown in Appendix Figure A4). Stability of the stock, however, hides a great deal of movement underneath. Tracking the linked pairs in both directions, Figure 3 shows that teaching loses and recruits between 15 and 20 percent of its stock every single year, so the profession is best pictured as a large revolving door whose two flows roughly balance, although in recent years exits have run persistently above entries.

Geography adds a final layer to the portrait, and here the interesting map is not where teachers live but where they quit. Teachers are a somewhat larger share of employment in New England and parts of the rural South, around 3.8 percent in Connecticut, Vermont or Mississippi, than in Nevada or Florida, at 2.2 percent, but attrition varies far more. As Figure 4 shows, annual exit rates range from about 11 percent in the most retentive states to above 19 in the least, and the variation lines up with what schools produce.

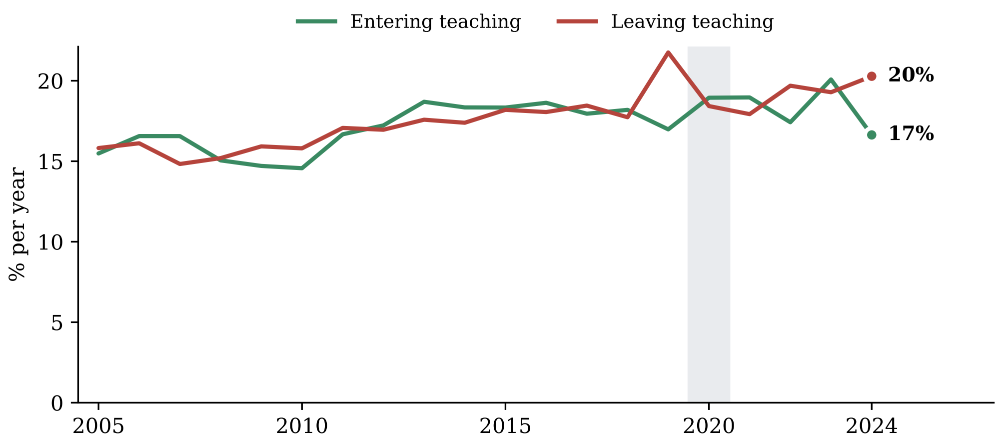
Linking each state to its official NAEP grade-8 mathematics score, the standard state measure of school quality, Figure 5 shows that states with better scores retain their teachers better, with a correlation of -0.47 and roughly one percentage point less attrition for every ten NAEP points. The direction of causality is of course open, since good schools may retain teachers and stable teaching forces may also produce good schools, but the association places teacher retention squarely inside the education-quality conversation.

4. Attrition over time

One in six 12-month leavers is teaching again within three months, 16.1 percent of women and 14.0 percent of men, and these temporary exits are precisely what the non-returning definition strips out. Levels in this section are the linked-design measure, the upper edge of the bracket that Appendix A reconciles with the official benchmarks, while the facts that carry the section, the trend and the composition of the flow, do not depend on where inside that bracket the true level sits. Net of that churn, Figure 7 shows that attrition net of returns still climbed from about 13 percent in 2005–2010 to about 17 percent in 2022–2024, peaking at 18.4 percent in the pandemic year and reaching a new high of 18.0 percent in 2024. Gender gaps are small and unstable throughout, and in the most recent year men, at 19.8 percent, actually exceeded women, at 17.4.

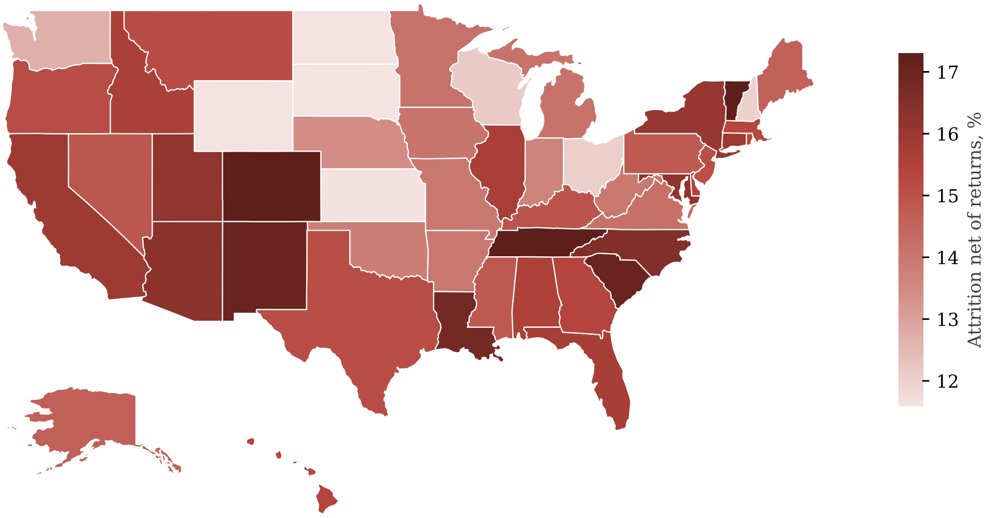
Is that level a lot? The question needs a benchmark, and a benchmark needs the same instrument on both sides, because the answer depends on the instrument as much as on the profession. When I apply the retrospective March question of Harris and Adams (2007) and Aldeman and Yi (2025) to my own CPS files, I reproduce their conclusions, with 6.7 percent of teachers, 7.5 percent of nurses, 5.6 percent of accountants and 15.9 percent of social workers leaving their occupation per year in 2022–2024, so under that instrument teacher turnover is unexceptional among degree-holding professions. The linked design measures the same exits at higher levels because it counts the job-to-job moves that a single recall question smooths away, and for professions with generic titles it also counts the churn of the occupational coders themselves, since accountants and social workers sit in dense code families where independent re-coding across interviews relabels many stayers (Mellow and Sider, 1983; Kambourov and Manovskii, 2013). Appendix Table A4 shows all four professions under both instruments. Figure 6 therefore draws

Figure 3: Teachers entering and leaving each year.



Notes: Gross annual flows in the linked pairs under the 12-month definitions. A person enters teaching if she is a degree-holding school teacher at the second interview but was not one a year earlier, and leaves in the reverse case. Because the linked sample misses people who move, both flows are conservative. Source: author's calculations from linked CPS pairs, 2005–2025.

Figure 4: Where teachers leave: attrition by state, 2005–2025.



Notes: Weighted non-returning attrition rate by state, pooling all twenty waves. Rates for the smallest states rest on fewer observations and should be read with care. Source: author's calculations from linked CPS pairs, 2005–2025.

the linked comparison where it is clean, teachers against registered nurses, the two female-dominated professions whose distinctive titles coders rarely miss, and counts only exits from the whole occupational field, so a promotion into school administration or a move from staff nurse to nurse practitioner is not an exit. Teaching is the stickier of the two in every single year, 13.8 percent per year against 17.4 on average, and the gap has not closed. Under either instrument, the notion of an exceptional teacher exodus finds no support.

5. Where leavers go

The linked design distinguishes those who abandon the labor force from those who change occupation, and for the latter it records the destination. Figure 8 shows that the single largest destination, at 30 percent of non-returning leavers, is another education job, whether postsecondary teaching, tutoring and instruction, teaching assistance, librarianship or school administration, so a large share of exits from classroom teaching never leave education at all. Retirement accounts for a further 17 percent, at a mean age of 62, and only 4 percent are unemployed a year later. The gender contrast is sharp exactly where the family results of the model will point, since 14 percent of women who leave are out of the labor force for reasons other than retirement or disability, at a mean age of 39 consistent with family care, against 5 percent of men, while men move more often into management and professional jobs, 26 against 19 percent.

The two margins have also moved differently over time. As Figure 9 makes clear, the rise in attrition documented in Section 4 is almost entirely a rise in occupation switching, from about 8.5 percent of teachers per year in the mid 2000s to 12 percent in 2022–2024, while exits into non-participation have stayed flat at 4 to 5 percent and unemployment spikes only in recessions. Teaching is increasingly losing people to other jobs, not to inactivity. Mobility inside the profession is substantial as well, since among teachers who are still teaching a year later 6.7 percent have moved from a public to a private school and

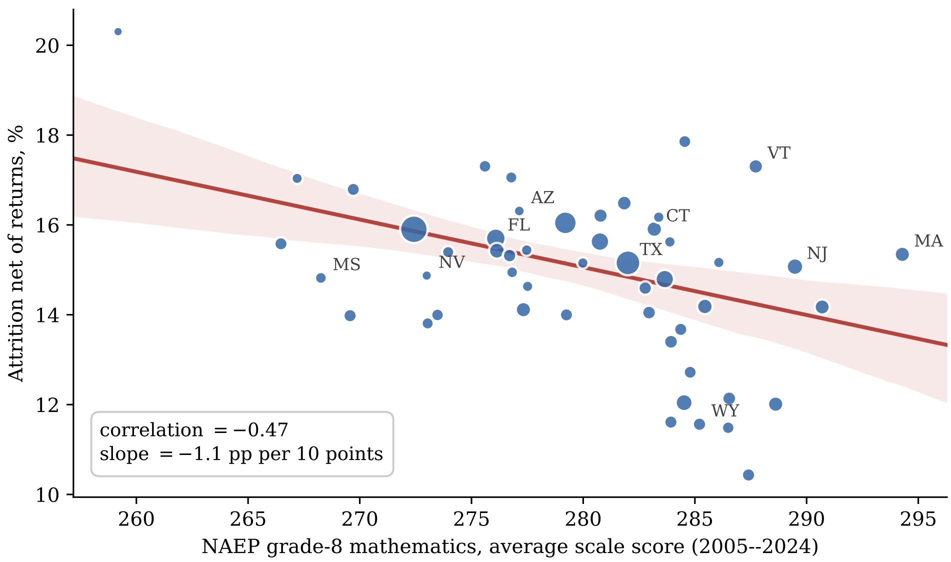
5.8 percent in the opposite direction, figures that class-of-worker coding noise makes upper bounds but that echo the school-to-school mover rates the follow-up surveys report.

Age reorganizes the same routes completely, as Figure 10 shows. Moving to another occupation is not a young teacher's phenomenon, it is the dominant route at every age below sixty, holding remarkably flat at 9 to 10 percent of teachers per year from the late twenties through the fifties and rising at both extremes. What changes over the lifecycle is everything else. Family and other labor-force exits fade with age, from 4.1 percent of teachers at ages 22 to 24 to 1.4 in the late forties, retirement is invisible before fifty and then takes over, reaching 13.6 percent per year at ages 60 to 64, and unemployment reaches one percent of teachers only at the very start of the career. A district worried about mid-career losses is worried, whether it knows it or not, about competition from other employers.

6. Who leaves

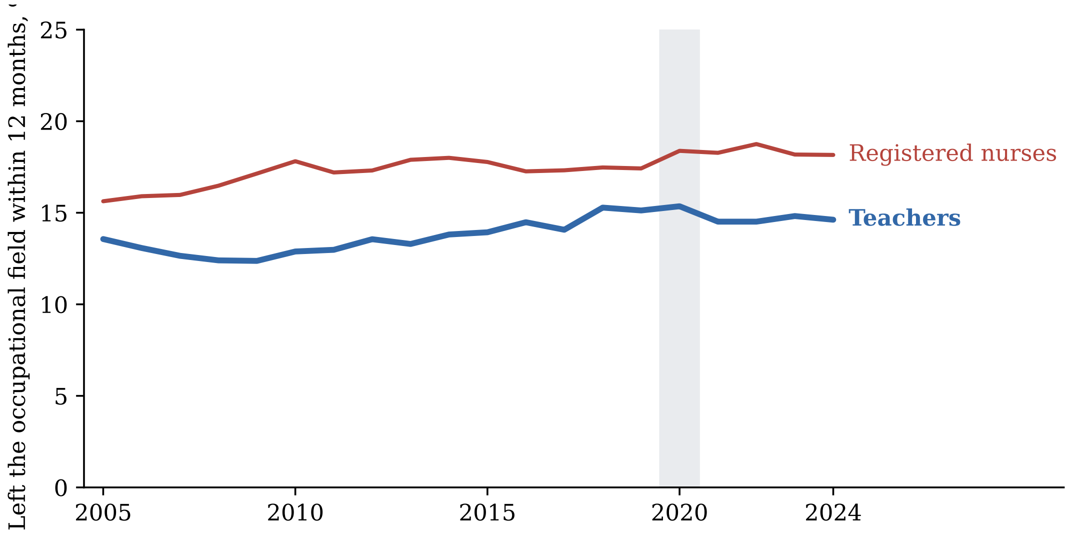
Appendix Table A6 opens the question by comparing stayers and non-returning leavers across every characteristic I observe, shading in green the differences that are both statistically significant and economically meaningful. Read as a whole, the table already sketches the profile of the leaver. Leavers are three years older on average, they are far more attached to the margins of the job than to its core, since working part-time is eleven points more common among them and private-sector employment fifteen points more common, and their family situation differs in a precise way, with fewer young children at home rather than more. The gender split, in contrast, is economically irrelevant, with women 76.8% of stayers and 77.0% of leavers, an early sign that gender operates through specific channels rather than through levels. Black and non-citizen teachers are clearly over-represented among leavers, and so are preschool and kindergarten teachers, while credentials matter in both directions, since leavers are less likely to hold a master's degree but almost twice as likely to hold a professional degree or doctorate. Most of these gaps are not marginal,

Figure 5: States with better NAEP scores lose fewer teachers.



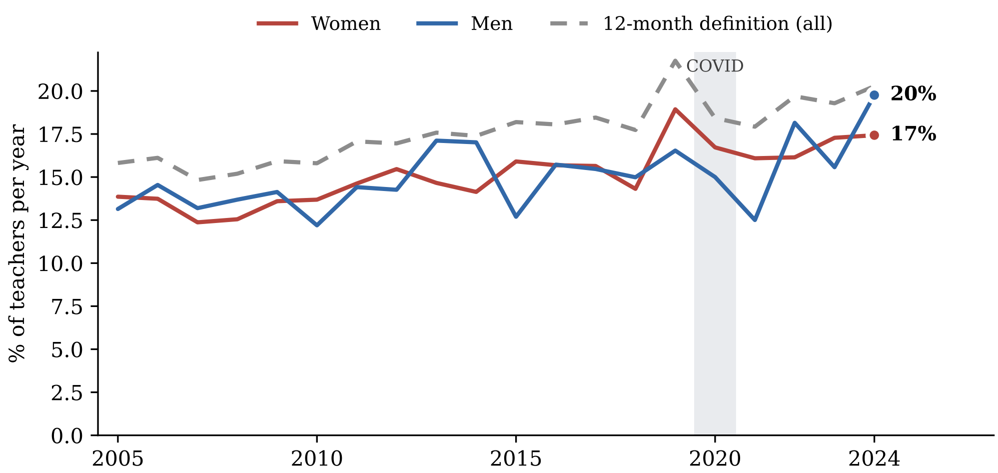
Notes: Each dot is a state, sized by its number of linked teachers. The horizontal axis averages the official NAEP grade-8 mathematics scale score over the 2005–2024 state assessments, retrieved from the Nation's Report Card data service; the vertical axis is the weighted non-returning attrition rate pooled over 2005–2025. The fitted line is weighted by sample size. Source: author's calculations from linked CPS pairs and NAEP.

Figure 6: Leaving the occupational field within twelve months, teachers and registered nurses under the same linked design.



Notes: Weighted shares of employed college graduates no longer employed in any occupation of their field twelve months later, education occupations for teachers and health-practitioner occupations for registered nurses, linked and validated exactly as the teacher panel and smoothed with a centered three-year moving average. Registered nurses are code 3130 before 2011 and codes 3255 to 3258 afterwards. Linked observations are 174,872 teachers and 75,160 nurses. The shaded band marks 2020. Appendix Table A4 reports all four professions under both the linked and the retrospective instruments. Source: author's calculations from linked CPS pairs, 2005–2025.

Figure 7: Attrition net of returns by year and gender.



Notes: The return window is capped by the survey design at three monthly interviews after the twelve-month link, and 21,873 of the 30,244 leavers are re-interviewed at least once. Source: author's calculations from linked CPS pairs, 2005–2025.

as ten of them exceed two percentage points, the threshold I use for economic relevance. Expressed as exit rates, part-time teachers leave at more than twice the rate of full-time teachers, 32 against 13 percent, and preschool and kindergarten teachers face 21 percent against 15 for elementary and 13 for secondary school.

The lifecycle shape is visible in the raw data as well. Plotting attrition by birth cohort separately for men and women in Figure 11 traces the same U that the model will estimate, with cohorts born before 1950, observed near retirement, leaving at 25 to 35 percent per year, mid-career cohorts at 11 to 13 percent, and the youngest climbing back to 18 percent. Men and women follow nearly the same curve, and the third line explains why the aggregate female curve hides the fertility margin. Among women with children at home, mid-life cohorts are the most stable group in the whole figure, a selection effect, while the youngest cohort of mothers leaves at almost 20 percent, the highest rate shown. Motherhood splits women into the most attached group and the most exit-prone one, and the average conceals both.

Splitting the age profile by decade shows where the rise of Section 4 lives. Figure 12 compares the U on the two halves of the sample, and the 2015–2024 curve sits above the 2005–2014 curve at every age, by 4.3 points at the entry ages of 22 to 24, by roughly two points through mid-career, and by three points at 60 to 64, while at ages 55 to 59 the two decades coincide. The rise in attrition is therefore not the aging of the workforce in disguise, since it is largest precisely at career entry, where teachers now leave at rates the previous decade reserved for workers fifteen years older.

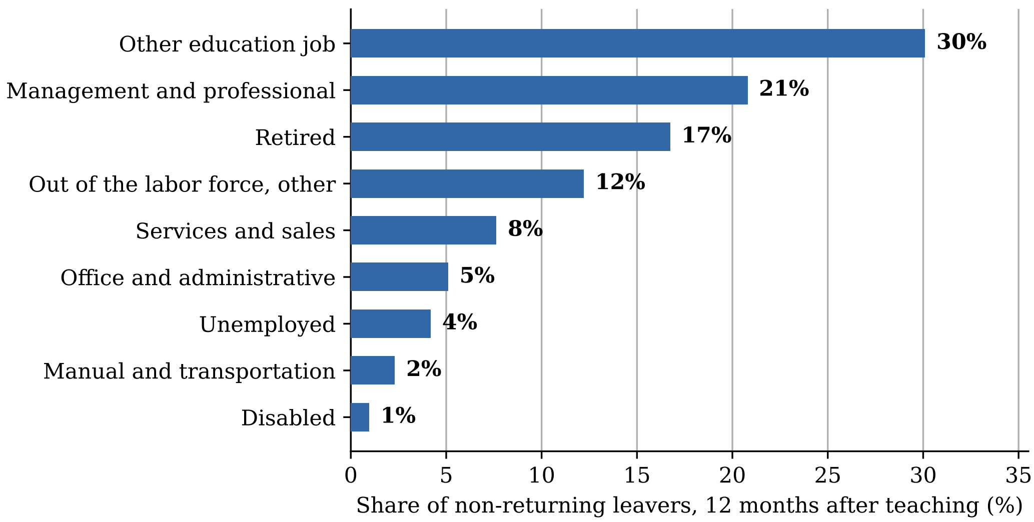
7. Why teachers leave

To weigh all these characteristics against each other at once, I estimate a probit model for the probability that teacher i, observed in year t, leaves without an observed return,

$$\Pr(\text{leave}_i = 1 \mid x_i) \;=\; \Phi\big(x_i'\beta + \gamma_t\big)\,, \tag{1}$$

where x_i collects everything I observe about the teacher at baseline, from demographics and family structure to hours, sector and region, γ_t are base-year fixed effects that absorb anything common to all teachers in a given year, such as the pandemic, and Φ is the standard normal distribution function that converts the index into a probability. In plain terms, the model asks how much more or less likely a teacher is to leave when one characteristic changes and all the others are held fixed, and I report each estimate as an average marginal effect, the change in the probability of leaving measured in percentage points. These are conditional associations in observational data, and none of them should be read as a causal effect; the model

Figure 8: Situation of non-returning leavers twelve months after teaching.



Notes: Education jobs other than school teaching include postsecondary teaching, tutoring and instruction, teaching assistance, librarians and education administration, and the remaining employed leavers are grouped into broad occupation classes. Retirement, disability, unemployment and other non-participation come from the labor force status recorded at the second interview. Source: author's calculations from linked CPS pairs, 2005–2025.

describes who leaves, holding everything else constant, not what would happen if a policy changed any of these traits. Standard errors are clustered by household. Appendix Table A9 reports the full estimates and Figure 13 plots the marginal effects, organized by the block each variable belongs to.

The job margin dominates. A part-time teacher is 11.3 percentage points more likely to leave than an otherwise identical full-time colleague (Appendix Table A9, column 2, significant at 1 percent), and a private-school teacher is 7.5 points more likely to leave than a public-school one. Against a baseline exit rate of 15 percent, the two largest effects in the model amount to roughly three quarters and one half of the average risk.

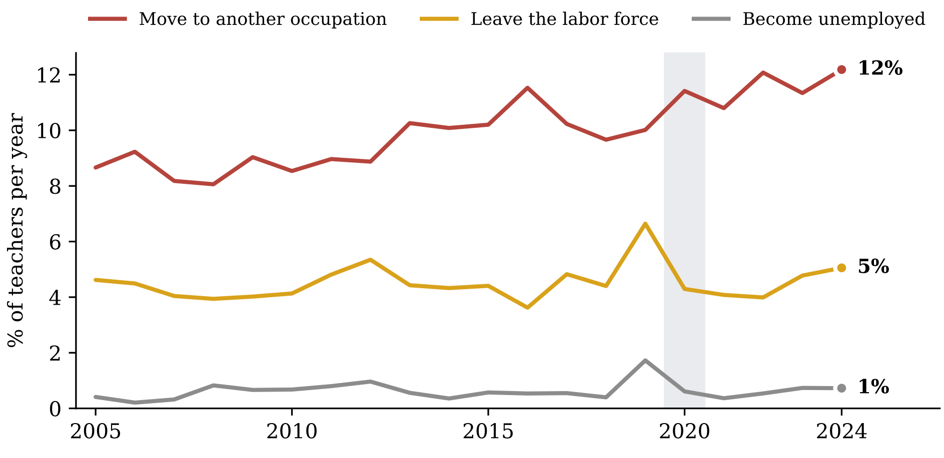
Pay retains, modestly. Weekly earnings are only collected in the outgoing rotations, so I estimate the wage effect on that subsample of 44,823 teachers, where ten percent higher weekly earnings lower the probability of leaving by 0.3 percentage points, about two percent of the baseline rate.

The family results separate stocks from events. The stock of children retains, at about one point less exit risk per own child at home (significant at 1 percent), while the event of a birth works in the opposite direction and only for women. A new baby during the year leaves men's exit probability unchanged, a precisely estimated zero with a p-value of 0.72, but raises women's by 5.1 points, significant at 5 percent and one third of the baseline risk, and 58 percent of the new mothers who leave exit the labor force entirely against 30 percent of the average leaver. This is the fertility channel the literature emphasizes (Stinebrickner, 2002), invisible when only the stock of children is used.

Minority and non-citizen teachers are significantly more likely to leave. Black teachers leave at 5.7 points above comparable white teachers and non-citizens at 5.7 points above citizens, both significant at 1 percent and each close to 40 percent of the baseline rate, while the Hispanic gap is a smaller 1.4 points, significant at 5 percent. These retention gaps compound into the diversity of the future workforce.

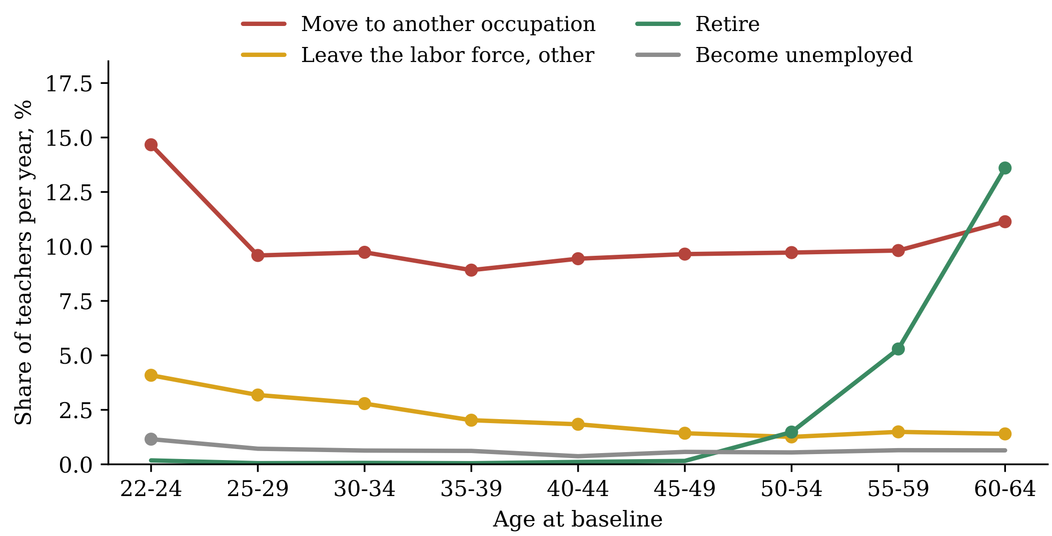
Credentials cut both ways, and age follows the familiar U. Relative to a bachelor's degree a master's retains modestly, while a professional degree or doctorate predicts 6.7 points more exit, significant at 1 percent and consistent with better options outside the classroom. The predicted probability of leaving falls to a minimum of 9 percent around age 39 and rises steeply after 55 (Appendix Figure A5). Every effect is nearly identical under the conventional 12-month definition, with the part-time effect at 13.6 points

Figure 9: Leavers by destination over time.



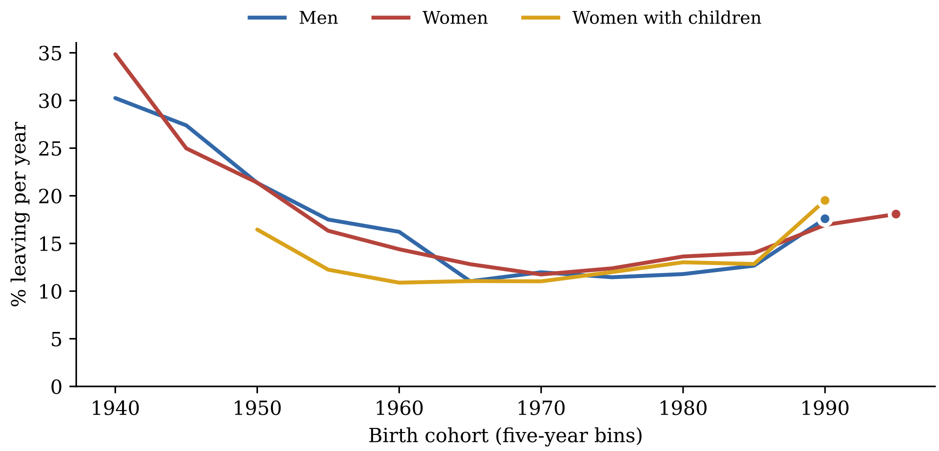
Notes: Non-returning leavers as a share of baseline teachers in each year, split by their situation twelve months later. Source: author's calculations from linked CPS pairs, 2005–2025.

Figure 10: Exit routes over the lifecycle.



Notes: Weighted share of teachers leaving through each route within twelve months, by age at baseline, pooling all 2005–2025 waves under the non-returning definition. Routes come from the labor force status recorded twelve months later; the labor-force route excludes retirement, shown separately, and includes disability. Markers are drawn only where a route reaches one percent of teachers per year. Source: author's calculations from linked CPS pairs, 2005–2025.

Figure 11: Attrition by birth cohort and sex.



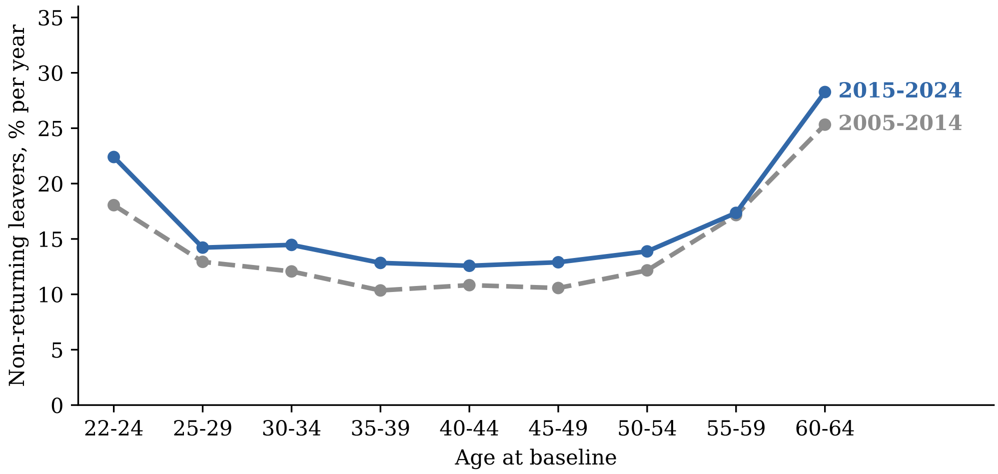
Notes: Weighted non-returning attrition rate by five-year birth cohort, pooling all twenty waves, so each cohort is observed at the ages it reaches during 2005–2025 and the curve mixes age and cohort variation. Women with children are those with at least one own child under eighteen at home at baseline, and cells with fewer than 400 observations are omitted. Source: author's calculations from linked CPS pairs, 2005–2025.

against 11.3, so the predictors do not depend on how leaving is measured.

None of this depends on the estimator. Re-estimating the same equation as a logit or as a linear probability model, or dropping the pandemic years 2019–2021 altogether, leaves every marginal effect essentially unchanged (Appendix Table A8); the part-time effect, for instance, moves between 10.8 and 14.7 points across the four variants and the female new-baby effect stays between 5.1 and 5.3.

Estimating the same model on subsamples, as Appendix Figure A6 does, confirms that the job effects are strikingly universal, with part-time work raising exit risk by 11 to 13 points for women and men, in public and private schools, below and above age 40, and the public-sector advantage at 7 to 8 points in

Figure 12: The age profile of exit, by decade.



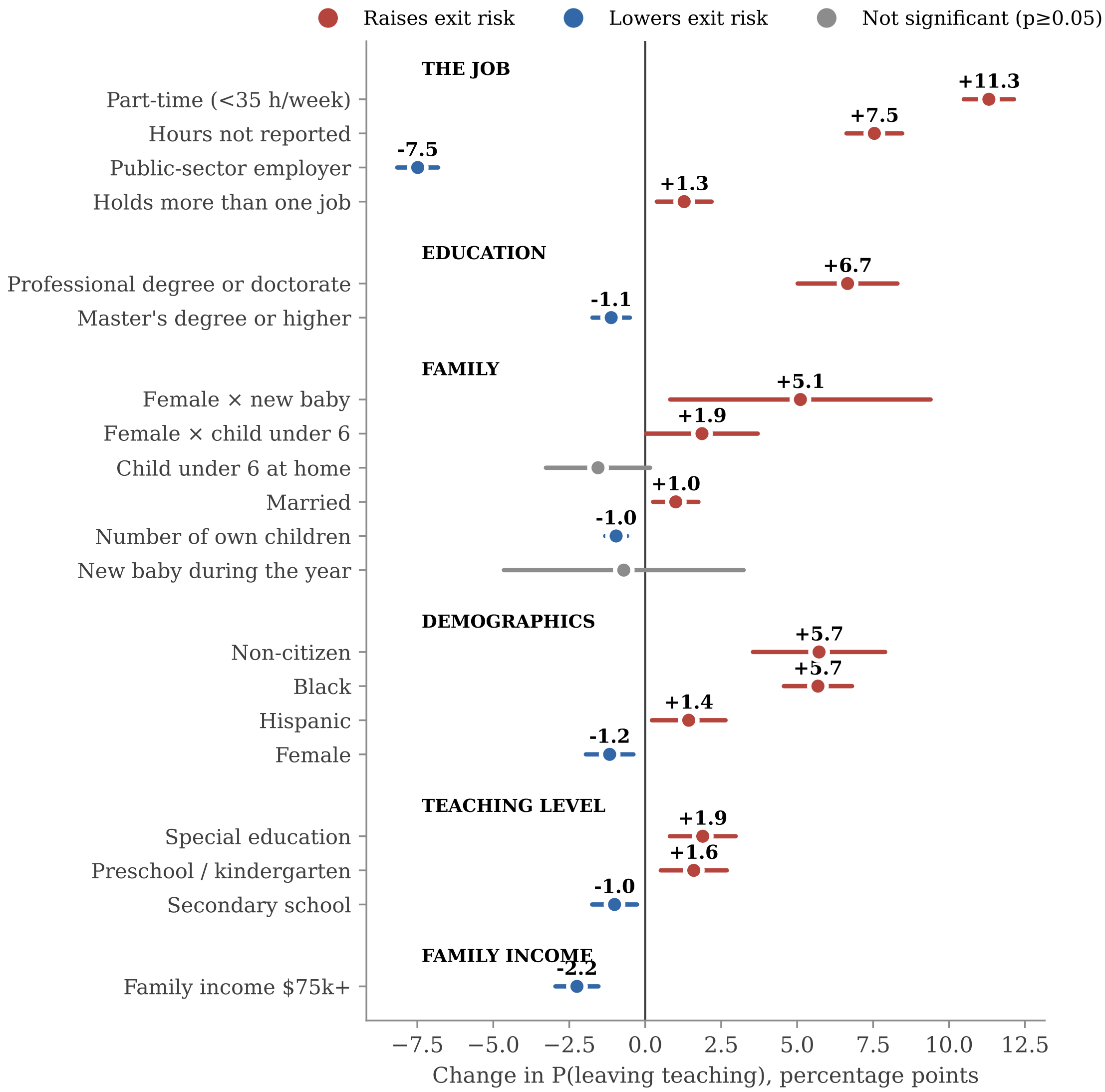
Notes: Weighted non-returning attrition rates by age at baseline, pooled within 2005–2014 and 2015–2024 baselines. The corresponding age profile estimated within the model, holding all other characteristics fixed, is in Appendix Figure A5. Source: author's calculations from linked CPS pairs, 2005–2025.

every cut. The fertility effect, in contrast, is entirely female, and the Black retention gap appears in every subsample, largest in private schools.

The routes tell more than the total. Taking to national data the question that Stinebrickner (2002) and Dolton and van der Klaauw (1999) could only ask in a small cohort and in British data, Appendix Table A7 re-estimates the model separately for each exit route, and the predictors split into two clean families. The opportunity variables work almost entirely through job switching, since the public-sector advantage is 6.4 of its 7.5 points on the occupation margin and a professional degree or doctorate raises switching by 6.3 points while leaving labor-force exit untouched. The family variables work almost entirely through non-participation, since a new baby moves mothers out of the labor force by 6.6 points while reducing their job switching by 2.3. Pay retains on both margins, and relative to the size of each flow it protects most where the flow is family driven.

Is teaching family friendly? Figure 14 runs the identical event for the three comparison professions, estimating the effect of a new baby on a woman's probability of leaving her occupational field with the same linked design, the profession-level counterpart of the child penalty literature (Kleven, Landais and Sogaard, 2019; Gulek, 2025). Teaching stands out, and not in the direction its reputation suggests. A new baby raises a teacher's probability of leaving the profession by 6.2 percentage points (standard error 0.9), the largest and by far the most precise effect of the four professions, while registered nurses show an effect indistinguishable from zero, and in every profession the baby suppresses job switching and operates only on the labor-force margin. New mothers step out of the labor force in nursing too, by 3.0 points, but nursing absorbs the shock without losing them from the field, while teaching converts nearly its entire labor-force exit of 6.9 points into a loss for the profession. The profession most often described as family friendly is the one that loses most new mothers, while nursing, where shift work makes hours genuinely divisible, loses none, the same part-time mechanism that Goldin and Katz (2016) document for pharmacy. Fathers show no effect in any profession.

Figure 13: Average marginal effects on the probability of leaving teaching, with 95% confidence intervals.



Notes: Estimates from the probit of Table A9. The age profile is shown in Appendix Figure A5. Region indicators and the hours-not-reported indicator are included in the model but not displayed.

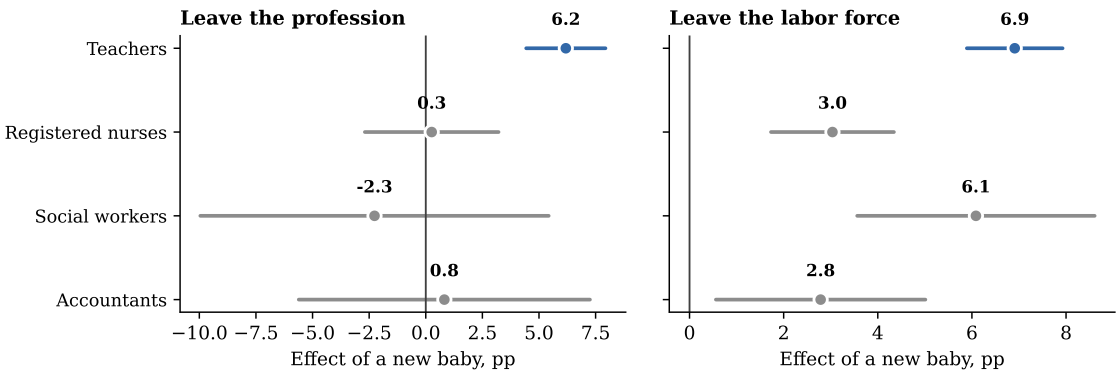
8. The typical leaver

Ranking teachers by their predicted risk gives the model's portrait of the typical leaver, drawn in Figure 15. The top decile faces a mean exit probability of 37 percent, 2.5 times the average teacher, and it is overwhelmingly female, at 79 percent, late-career with a mean age of 55, six times more likely to work part-time, 56 against 9 percent, and much less anchored in the public sector, where only 40 percent of the group works against 78 percent of all teachers, with Black teachers over-represented at 14 against 6 percent.

9. Policy implications

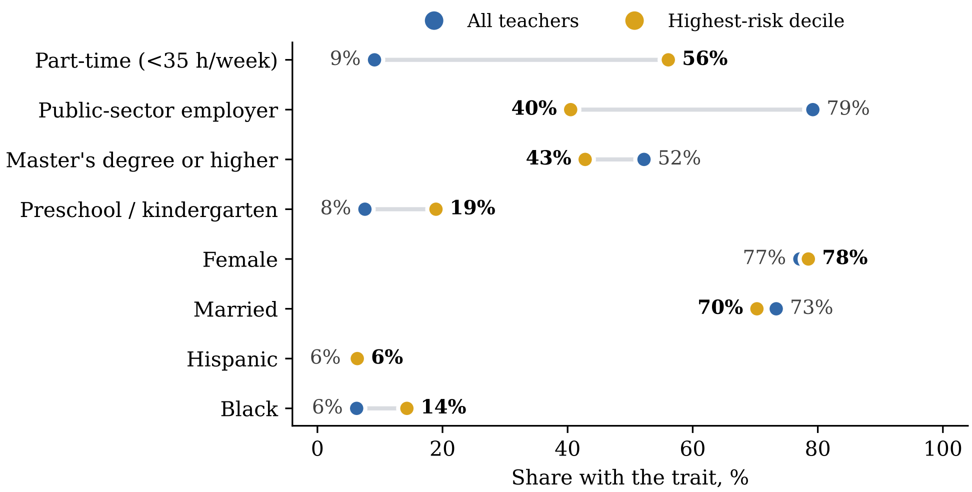
The results locate attrition in weak attachment to the job, in part-time hours, private-sector positions and both career extremes, rather than in broad demographics, with a rising trend made almost entirely of moves to other occupations. Four levers follow directly. Converting involuntary part-time positions to

Figure 14: A new baby and the probability that a woman leaves her profession.



Notes: Average marginal effects of a new baby, an own child under three appearing between the two interviews, from probits estimated on women of each profession with age, age squared, marital status, an advanced-degree indicator and base-year fixed effects, with 95% confidence intervals and standard errors clustered by household. Leaving the profession is leaving the whole occupational field under the conventional 12-month definition, so a move inside education or inside health practice is not an exit; the teacher effect of 6.2 points sits next to the 5.1 of the non-returning interaction in Appendix Table A9. New-baby events are 3,212 for teachers, 1,599 for nurses, 636 for accountants and 415 for social workers. Source: author's calculations from linked CPS pairs, 2005–2025.

Figure 15: Characteristics of the top predicted-risk decile vs all teachers.



Notes: Predicted probabilities from the probit of Table A9. Source: author's calculations from linked CPS pairs, 2005–2025.

full-time attacks the single largest risk factor, and the wage result implies that pay raises retain, though modestly. The private sector is where teaching careers break, and part of the preschool exposure reflects that composition. The Black, Hispanic and non-citizen retention gaps deserve monitoring for the diversity of the future workforce. And with one in six leavers back within three months, cheap re-entry pathways such as fast-track rehiring and return bonuses can recover part of the outflow at low cost.

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Appendix A. Sample construction and representativeness

This appendix documents how the analytic sample is built from the raw survey, how representative it remains after linkage, and the measurement limits a reader should keep in mind.

Table A1 traces how the CPS universe narrows down to my analytic sample. In an average month the survey interviews close to 95,000 adults who represent 241 million people, of whom about 1,850 are employed school teachers with at least a bachelor's degree, representing the 4.6 million teachers of the country. The degree requirement removes a small group of daycare and substitute personnel. Only the half of the sample that is in its first rotation block can be linked one year ahead, and the linkage itself succeeds for 78 percent of teachers. Reassuringly, it succeeds less often for everyone else, 74 percent among other college-educated workers and 71 percent among all employed, a ranking that holds in every single year, so sample losses do not select against teachers. Losses arise because the CPS samples addresses rather than people, which means movers drop out, and since moving is correlated with changing jobs my attrition rates are conservative on this margin. Linked persons keep their population weights, so all weighted rates remain representative of the 4.6 million teachers.

Table A1: From the CPS universe to the analytic sample.

	Persons	Population represented
Adults interviewed in an average month	94,630	240.6M
employed	57,572	148.1M
school teachers (occ. 2300–2330)	2,094	5.2M
with bachelor's degree or higher	1,849	4.6M
Linked 12 months later (54,944 teachers, 20 waves)	174,872	the same 4.6M
with re-interviews after t+12: **main sample**	129,897	the same 4.6M

Notes: Monthly figures are averages over the 251 available months, with weighted population totals beside the sample counts. Teachers are employed persons whose main job is school teaching (Census occupation codes 2300 to 2330) holding a bachelor's degree or higher. Source: author's calculations from CPS basic monthly microdata, 2005–2025.

Two measurement caveats apply throughout. First, my observation window is genuinely short. A person leaves the survey for good at most sixteen months after entering it, so a teacher who returns to the classroom more than a few months after the twelve-month link is still counted as a leaver, and all my attrition rates are upper bounds on permanent exit. Measuring multi-year exits and returns would require a long panel such as the SIPP or the NLSY. Second, occupation in the returning rotation is re-coded independently, which mechanically inflates measured switching, so levels should again be read as upper bounds while comparisons across groups and over time, the object of this paper, are far less affected.

How do these rates compare with the published benchmarks? Table A2 walks the distance step by step, and every step is an institutional difference between the sources rather than an adjustment. The Teacher Follow-up Survey samples K–12 classroom teachers from school rosters, explicitly excluding short-term substitutes, student teachers and teacher aides, and reports that 8.0 percent of public school teachers left the classroom between 2020–21 and 2021–22 (NCES, 2024). My universe is instead anyone whose main job is school teaching, substitutes included. Purging observed returns lowers my conventional 17.6 percent to 15.1, restricting to public K–12 teachers lowers it to 12.7, dropping part-timers, the closest observable proxy for the substitutes that the rosters exclude, lowers it to 11.5, and counting as leavers only teachers found in no education occupation at t+12, the margin where independent occupation coding is most fragile, yields 8.0. This last step deliberately overshoots, because the follow-up survey counts moves into non-teaching education jobs as leaving while my strictest row does not, so the fully aligned figure lies between 8.0 and 11.5 and the survey benchmark sits inside that bracket. The private side tells the same story from the other direction, since the follow-up survey itself reports 12.0 percent attrition in private schools, and its leavers are not gone from education either, as 39 percent of them keep working for a school or district in a non-teaching role, the same phenomenon as the 30 percent of education destinations in Figure 8.

The retrospective CPS estimates sit below all of this, and I can reproduce them exactly. Applying the definition of Harris and Adams (2007) to the 2022–2024 March supplements, a teacher by the longest job of the previous calendar year and a leaver by the survey-week occupation, yields 5.8 percent with my bachelor's restriction and 6.7 percent without it, against the 7.6 percent that Aldeman and Yi (2025) obtain pooling 2015–2024. The same survey therefore delivers 17.6 percent when interviews a year apart are linked and 6 to 7 percent when one interview asks about the year before, and the decomposition shows where the signal goes, since the retrospective instrument records only 1.9 percentage points of moves into other occupations against about 12 in the linked pairs, while the non-employment margin is similar in both. Recall and dependent coding compress precisely the job-to-job margin that drives my series. The address-based design also means I follow dwellings rather than people, and indeed not a single continuing teacher in my linked sample changes state, so cross-district and cross-state moves are invisible here and my rates are conservative on that margin.

Linkage also preserves composition. Table A3 compares the weighted profile of my linked baseline with the universe of all teacher interviews over the same months. Age differs by one year and every share by less than two percentage points, with the panel tilted slightly toward the older, settled teachers who do not change address, which is the conservative direction for attrition.

Table A2: From this paper's definitions to the published benchmarks.

	Annual rate	Observations
Conventional 12-month leaver, all teachers	17.6%	174,872
Non-returning leaver (main definition)	15.1%	129,897
public K–12 teachers only	12.7%	98,122
and full-time only	11.5%	91,035
and in no education occupation at $t{+}12$	8.0%	91,035
Benchmarks and retrospective measures		
Teacher Follow-up Survey, public school leavers	8.0%	
Teacher Follow-up Survey, private school leavers	12.0%	
My retrospective replication, 2022–2024 ASEC	5.8%	6,945
without the degree restriction	6.7%	7,909
Harris and Adams (2007); Aldeman and Yi (2025)	7.7; 7.6%	

Notes: Weighted annual rates. The ladder starts from the conventional 12-month definition on all linked teachers and then applies the non-returning definition on the baseline subsample whose rotation allows re-interviews, restricting it step by step to the Teacher Follow-up Survey universe; full-time status is the observable proxy for the roster-based exclusion of short-term substitutes, and the final row counts as leavers only teachers found in no education occupation twelve months after baseline. The benchmark rows quote the 2021–22 Teacher Follow-up Survey leaver rates (NCES, 2024); the replication rows apply the retrospective definition of Harris and Adams (2007) to the 2022–2024 March supplements, with teachers identified by the longest job of the previous calendar year and leaving defined by the survey-week occupation. Source: author's calculations from linked CPS pairs and March supplements, 2005–2025.

Table A3: Composition of the linked panel vs all teacher interviews.

	All teacher interviews	Linked panel
Age, years	43.0	44.0
Female	76.6%	76.8%
White	86.2%	87.6%
Master's degree or higher	51.2%	52.9%
Public-sector employer	76.8%	78.3%
Part-time (<35 h/week)	8.7%	9.0%
Unique persons	98,972	54,944

Notes: Weighted means. The first column pools every monthly interview of an employed, degree-holding school teacher between 2005 and 2025; the second column is the baseline of the linked 12-month panel used throughout the paper. Source: author's calculations from CPS basic monthly microdata, 2005–2025.

Table A4: Four professions under the linked and the retrospective instruments.

	Linked design, 2005–2024			Retrospective, 2022–2024	
	Left detailed occupation	Left occupational field	Obser-vations	Left detailed occupation	Sample
Teachers	17.4	13.8	174,872	6.7	7,909
Registered nurses	20.2	17.4	75,160	7.5	5,176
Social workers	41.3	33.3	22,874	15.9	1,170
Accountants and auditors	38.8	32.4	49,673	5.6	2,088

Notes: Annual exit rates in percent, weighted. The linked columns come from 12-month occupation pairs built, linked and validated exactly as the teacher panel, pooled over 2005–2024 baselines, under the conventional definition; field exit counts only moves outside the whole occupational field, education occupations for teachers, health-practitioner occupations for nurses, community and social service occupations for social workers and business and financial operations occupations for accountants. The retrospective columns replicate the design of Harris and Adams (2007) and Aldeman and Yi (2025) on the 2022–2024 March supplements, comparing the occupation of the longest job held the previous calendar year with the survey-week occupation, without a degree restriction, weighted by supplement weights; Aldeman and Yi report a 7.6 percent rate for teachers pooling 2015–2024. The gap between instruments reflects the job-to-job moves that a single recall question smooths away and, for the generic titles of accountants and social workers, the independent re-coding of detailed occupations across interviews (Mellow and Sider, 1983; Kambourov and Manovskii, 2013), which is why the body of the paper draws the linked comparison only between teachers and nurses. Source: author's calculations from linked CPS pairs and March supplements.

Appendix B. Additional figures and tables

This appendix collects the supporting material referenced in the main text, moving from description to estimation. It shows the age and composition contrasts behind Table 1, the stock of teachers and its composition by teaching level, the sub-period profile of the workforce and the stayer-leaver comparison, and then the model material, namely the predicted age profile, the subsample estimates, the estimator robustness checks and the full regression output.

Table A5: The changing profile of the teaching workforce, by sub-period.

	2005–2011	2012–2018	2019–2025	Change
Age, years	42.3	42.8	42.9	0.63***
Under 35	31.7%	30.4%	29.1%	-2.6***
Female	78.3%	78.3%	78.0%	-0.3
White	86.7%	84.9%	83.1%	-3.7***
Hispanic	7.2%	9.7%	11.2%	4.0***
Married	68.0%	66.9%	64.8%	-3.2***
Child under 6 at home	18.3%	18.0%	15.9%	-2.4***
Bachelor's degree or higher	87.5%	87.3%	89.1%	1.6***
Master's degree or higher	41.9%	45.3%	47.9%	6.0***
Public-sector employer	74.8%	71.4%	70.3%	-4.5***
Union member	48.0%	44.6%	43.3%	-4.6***
Part-time	11.2%	11.4%	9.4%	-1.8***
Persons	43,697	44,405	35,890	

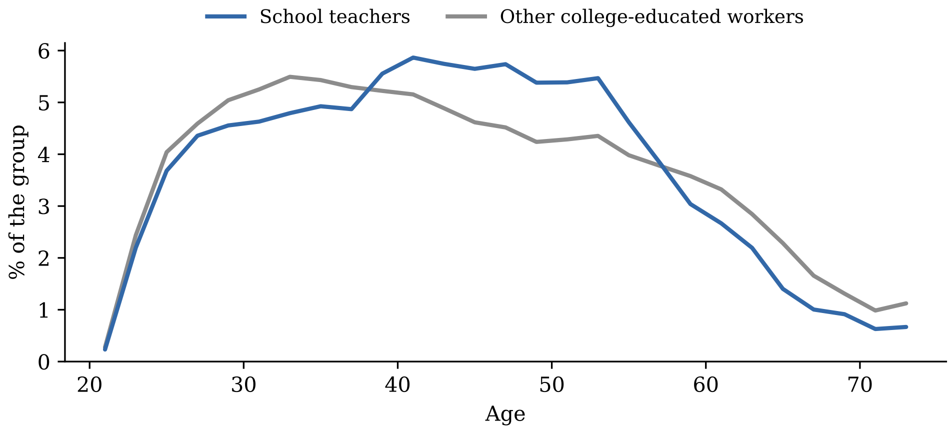
Notes: Weighted means over all employed persons observed in a school-teaching occupation, with no degree restriction, in each sub-period. Sample sizes are unique persons. The last column reports the change between 2005–2011 and 2019–2025, in percentage points for shares, and the significance of a two-sample test with standard errors clustered by household ($^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$). Union membership is only asked in the outgoing rotations. Source: author's calculations from CPS basic monthly microdata, 2005–2025.

Table A6: Stayers vs non-returning leavers in the analytic sample (weighted means).

	All teachers	Stayers	Non-returning leavers	Difference
Age, years	44.0	43.5	46.7	3.15***
Female	76.8%	76.8%	77.0%	0.2
Female aged 25–44	37.8%	38.8%	31.7%	-7.1***
Married	72.0%	72.5%	69.5%	-3.0***
Number of own children (<18) at home	0.8	0.9	0.6	-0.22***
Child under 6 at home	18.1%	18.9%	13.9%	-5.0***
New baby during the year	2.5%	2.5%	2.8%	0.3
Black	8.1%	7.5%	11.6%	4.2***
Hispanic	7.9%	7.8%	8.5%	0.7*
Non-citizen	1.7%	1.5%	3.0%	1.5***
Master's degree or higher	52.9%	53.7%	48.4%	-5.3***
Professional degree or doctorate	2.8%	2.5%	4.4%	2.0***
Part-time (<35 h/week)	9.0%	7.2%	19.1%	11.9***
Holds more than one job	8.1%	7.9%	9.4%	1.5***
Public-sector employer	78.3%	80.5%	65.7%	-14.9***
Family income $75k+	76.4%	77.3%	71.4%	-5.9***
Elementary / middle school	61.8%	61.9%	61.2%	-0.7
Preschool / kindergarten	7.7%	7.1%	10.6%	3.5***
Secondary school	23.1%	23.5%	20.5%	-3.0***
Special education	7.5%	7.5%	7.6%	0.1
Observations	129,897	110,720	19,177	

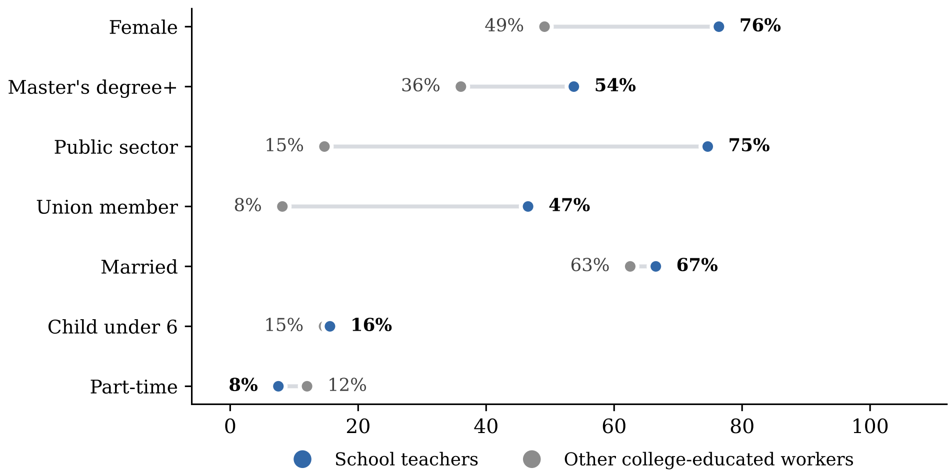
Notes: The sample is all linked teachers whose rotation includes at least one interview after the twelve-month link. All variables are measured at baseline and weighted with CPS person weights. Children are own children under eighteen living in the household at the time of the interview. The last column reports the stayer-leaver difference, in percentage points for shares, and the significance of a two-sample test with standard errors clustered by household ($^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$); green cells mark differences that are significant at 5 percent and larger than two percentage points (one year of age, 0.15 children). Source: author's calculations from CPS basic monthly microdata, 2005–2025.

Figure A1: Age distribution of school teachers and of other college-educated workers, 2021–2025.



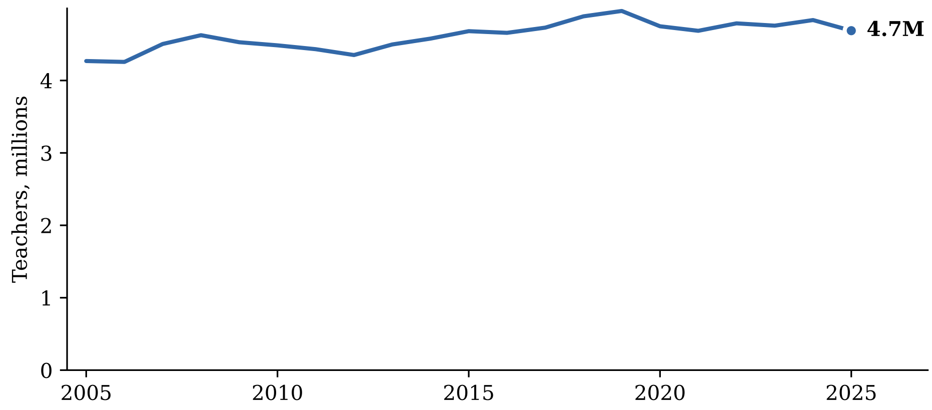
Notes: Both groups are employed persons with at least a bachelor's degree, and teachers are identified as in Table A1. Weighted to population. Source: author's calculations from CPS basic monthly microdata, 2021–2025.

Figure A2: Composition of the teaching workforce vs other college-educated workers, 2021–2025.



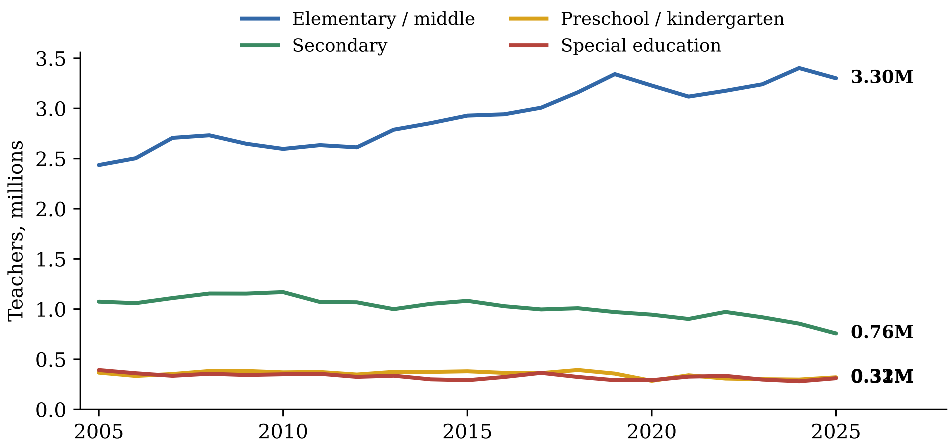
Notes: Same groups as Figure A1. Union membership is only asked in the outgoing rotations. Shares are weighted to population. Source: author's calculations from CPS basic monthly microdata, 2021–2025.

Figure A3: The stock of school teachers.



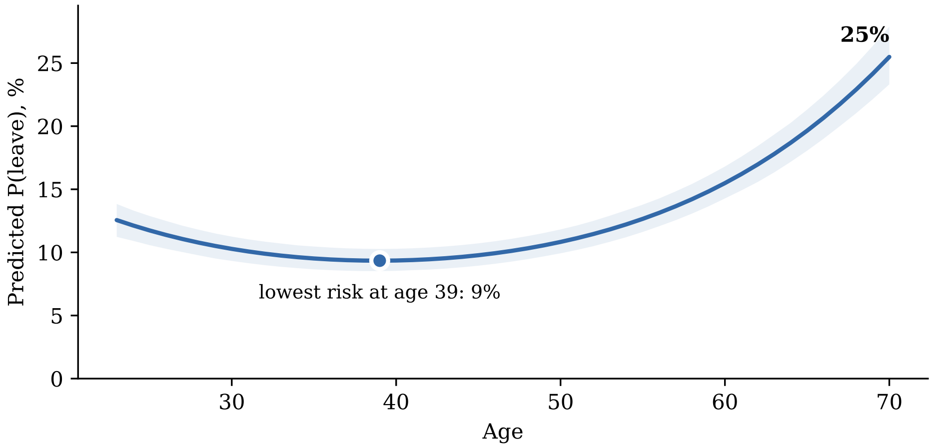
Notes: Employed school teachers with a bachelor's degree or higher, annual average of monthly weighted estimates. Source: author's calculations from CPS basic monthly microdata, 2005–2025.

Figure A4: Teachers by teaching level.



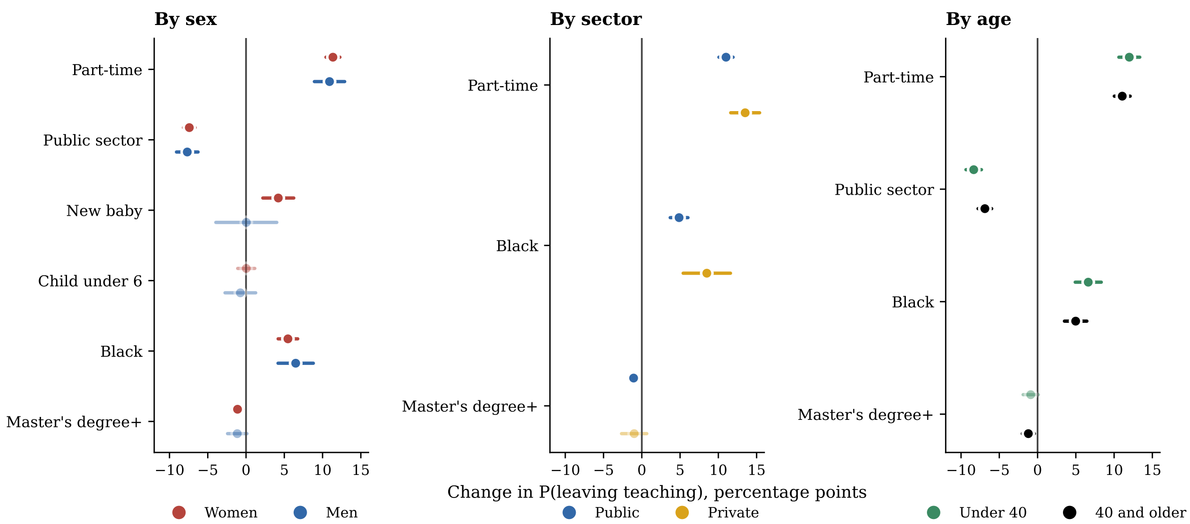
Notes: Same identification as Figure A3, split by occupation code. Source: author's calculations from CPS basic monthly microdata, 2005–2025.

Figure A5: Predicted probability of leaving by age.



Notes: Predicted probabilities from the probit of Table A9, holding all other covariates at their sample means, with a simulation-based 95% confidence band. Source: author's calculations from linked CPS pairs, 2005–2025.

Figure A6: The same model estimated on subsamples.



Notes: Average marginal effects with 95% confidence intervals from the probit of Table A9 estimated separately on each subsample; lighter marks are not significant at the 5 percent level. The birth and young-child effects are shown only in the sex panel because they are sex-specific. Source: author's calculations from linked CPS pairs, 2005–2025.

Table A7: Exit routes. Probit average marginal effects by destination.

	(1) Any exit	(2) Another occupation	(3) Out of the labor force	(4) Unemployed
Part-time (<35 h/week)	11.31***	6.97***	3.47***	0.55***
	(0.42)	(0.35)	(0.22)	(0.07)
Public-sector employer	-7.49***	-6.40***	-0.65***	-0.27***
	(0.34)	(0.28)	(0.19)	(0.06)
Professional degree or doctorate	6.66***	6.34***	-0.18	0.17
	(0.84)	(0.66)	(0.48)	(0.15)
Master's degree or higher	-1.12***	-0.68**	-0.42**	-0.16***
	(0.31)	(0.26)	(0.17)	(0.06)
Black	5.68***	3.91***	1.25***	0.34***
	(0.57)	(0.47)	(0.31)	(0.09)
Non-citizen	5.72***	3.08***	1.88***	0.48***
	(1.11)	(0.90)	(0.56)	(0.15)
New baby during the year	-0.71	-0.58	-0.83	-0.13
	(2.01)	(1.62)	(1.51)	(0.40)
Female × new baby	5.11**	-2.33	6.58***	0.02
	(2.19)	(1.83)	(1.57)	(0.45)
Number of own children	-0.96***	-0.41***	-0.81***	-0.11***
	(0.18)	(0.15)	(0.12)	(0.04)
Female	-1.17***	-1.63***	0.72***	-0.21***
	(0.40)	(0.33)	(0.22)	(0.07)
Outgoing-rotation subsample with weekly earnings				
Log weekly earnings	-3.18***	-1.20***	-1.28***	-0.33***
	(0.33)	(0.27)	(0.18)	(0.08)
Base-year fixed effects	Yes	Yes	Yes	Yes
Full covariate set	Yes	Yes	Yes	Yes
Mean of dependent variable	0.148	0.097	0.045	0.006
Observations	129,897	129,897	129,897	129,897
wage subsample	44,823	44,823	44,823	44,823

Notes: Column (1) reproduces the main model; columns (2) to (4) estimate the same specification with the dependent variable equal to one for each exit route, which partition the non-returning leaver. Average marginal effects in percentage points with delta-method standard errors clustered by household in parentheses; estimation is unweighted. The earnings row comes from the outgoing-rotation subsample, where the re-interview window does not exist, so it uses the conventional 12-month definition. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A8: Robustness to estimator and sample. Dependent variable equals one for non-returning leavers.

	(1) Probit (baseline)	(2) Logit	(3) Linear probability model	(4) Probit, excl. 2019–2021
Part-time (<35 h/week)	11.31***	10.78***	14.70***	10.75***
	(0.42)	(0.40)	(0.63)	(0.45)
Public-sector employer	-7.49***	-7.33***	-8.43***	-7.50***
	(0.34)	(0.34)	(0.43)	(0.37)
Professional degree or doctorate	6.66***	6.48***	7.89***	6.38***
	(0.84)	(0.80)	(1.12)	(0.90)
Master's degree or higher	-1.12***	-1.18***	-0.99***	-1.12***
	(0.31)	(0.32)	(0.31)	(0.33)
Black	5.68***	5.60***	6.42***	5.57***
	(0.57)	(0.56)	(0.72)	(0.61)
Non-citizen	5.72***	5.53***	7.26***	5.76***
	(1.11)	(1.05)	(1.54)	(1.18)
New baby during the year	-0.71	-0.91	-0.53	-0.51
	(2.01)	(2.13)	(1.70)	(2.11)
Female × new baby	5.11**	5.31**	5.18***	5.15**
	(2.19)	(2.28)	(1.97)	(2.29)
Number of own children	-0.96***	-1.04***	-0.65***	-0.87***
	(0.18)	(0.19)	(0.17)	(0.20)
Family income $75k+	-2.25***	-2.21***	-2.37***	-2.26***
	(0.36)	(0.36)	(0.39)	(0.38)
Female	-1.17***	-1.17***	-1.24***	-1.66***
	(0.40)	(0.40)	(0.42)	(0.42)
Base-year fixed effects	Yes	Yes	Yes	Yes
Full covariate set	Yes	Yes	Yes	Yes
Mean of dependent variable	0.148	0.148	0.148	0.145
Observations	129,897	129,897	129,897	113,751
Unique teachers	52,238	52,238	52,238	45,816

Notes: Each column re-estimates the full specification of Appendix Table A9 on the same analytic sample. Columns (1), (2) and (4) report average marginal effects in percentage points from probit and logit models; column (3) reports linear probability model coefficients multiplied by one hundred, so all four columns are directly comparable. Column (4) drops the 2019–2021 base years. All models are estimated without survey weights and include base-year fixed effects; delta-method standard errors clustered by household in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A9: Determinants of leaving teaching. Probit, dependent variable equals one for non-returning leavers, 2005–2025.

	Leaves teaching with no observed return	
	(1) Probit coefficients	(2) Marginal effect (percentage points)
Age (years)	-0.053***	-1.14***
	(0.004)	(0.09)
$\text{Age}^2/100$	0.068***	1.46***
	(0.004)	(0.09)
Female	-0.055***	-1.17***
	(0.019)	(0.40)
Married	0.047***	1.01***
	(0.018)	(0.38)
Number of own children	-0.045***	-0.96***
	(0.009)	(0.18)
Child under 6 at home	-0.072*	-1.55*
	(0.041)	(0.87)
Female × child under 6	0.087**	1.87**
	(0.044)	(0.94)
New baby during the year	-0.033	-0.71
	(0.094)	(2.01)
Female × new baby	0.238**	5.11**
	(0.102)	(2.19)
Black	0.265***	5.68***
	(0.027)	(0.57)
Hispanic	0.067**	1.43**
	(0.029)	(0.62)
Non-citizen	0.267***	5.72***
	(0.052)	(1.11)
Master's degree or higher	-0.052***	-1.12***
	(0.015)	(0.31)
Professional degree or doctorate	0.311***	6.66***
	(0.039)	(0.84)
Part-time (<35 h/week)	0.528***	11.31***
	(0.020)	(0.42)
Hours not reported	0.352***	7.54***
	(0.022)	(0.47)
Holds more than one job	0.060***	1.28***
	(0.021)	(0.46)
Public-sector employer	-0.349***	-7.49***
	(0.016)	(0.34)
Family income $75k+	-0.105***	-2.25***
	(0.017)	(0.36)
Midwest	-0.082***	-1.76***
	(0.021)	(0.45)
South	0.007	0.15
	(0.020)	(0.42)
West	-0.017	-0.36
	(0.021)	(0.45)
Preschool / kindergarten	0.075***	1.60***
	(0.026)	(0.55)
Secondary school	-0.047***	-1.01***
	(0.018)	(0.38)
Special education	0.088***	1.89***
	(0.026)	(0.55)
Base-year fixed effects	Yes	Yes
Mean of dependent variable	0.148	0.148
Observations	129,897	129,897
Unique teachers	52,238	52,238
Pseudo R^2	0.068	

Notes: The outcome is the non-returning leaver defined in Section 2. Column (1) reports probit coefficients and column (2) the corresponding average marginal effects in percentage points, with standard errors clustered by household in parentheses below each estimate, delta-method in column (2). The sample is all school teachers with a bachelor's degree or higher whose rotation includes at least one interview after the twelve-month link; the model is estimated without survey weights. Reference categories are full-time hours, a bachelor's degree, a private-sector employer, elementary or middle school, and the Northeast region. These estimates are conditional associations, not causal effects. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.